Anonymity as an Aggregation Condition:

Governance, Resource Structure, and Membership Dynamics in Twelve-Step Mutual-Aid Organizations

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October 2026

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Abstract

Alcoholics Anonymous (AA) has operated for more than ninety years as a worldwide fellowship of autonomous groups under a written constitution, the Twelve Traditions, that has not been substantially revised. AA keeps no membership lists, and the edition of its published membership table read here stops at 2020, so no present count is asserted here. This paper develops three connected formal results and reports the robustness analysis that determines how much each of them is worth.

First, and centrally, three of the Traditions (group conscience with servant leadership, the refusal to organize hierarchically, and anonymity with principles placed before personalities) are read as jointly enforcing the condition that maximum influence weight within a group vanishes as the group grows, which Golub and Jackson (2010) prove is necessary and sufficient for naive opinion averaging to aggregate dispersed private information. If the reading is right, AA's governance rules implement a known aggregation criterion, reached inductively in 1946, six decades before its formal statement. *The reading is the paper's largest unverified step and is flagged as such wherever it is used.* Two corollaries not stated in the Traditions follow: rotation of service satisfies the criterion only if the rotation pool scales with group size, and whether a concentration of attention obstructs learning at all depends on how it scales with the group rather than on how severe it looks at any one size.

Second, we derive the coupling between the Twelve Steps and Twelve Traditions through an intermediate layer of group-produced resources, rejecting the natural conjecture that the two lists pair by index and recovering, without assuming them, a two-tier functional division among the Traditions and the primacy of unity that Tradition 1 asserts of itself. We then report, against our own interest, that five of the twelve rejections are structurally forced by sparsity and invisible to multiplicative perturbation, that a threshold design able to see them finds them failing by comparable margins, and that the central coupling claim survives structural randomization in only 40.6 per cent of draws, which is less often than it fails. The coupling results rest on the magnitudes of two hand-written matrices and are argued for rather than certified.

Third, a stochastic simulation with endogenous membership, respecting the institutional fact that an AA group cannot refuse *membership*, distinguishes two channels of newcomer supply and shows that groups starved of exogenous referrals decline demographically while per-member quality among survivors is maintained or rises, whereas groups with exclusionary internal culture shrink, close outright in one run in four, and remain viable in a little over half. Sensitivity analysis over all 118 registered numeric values uses 1,002 global draws, 2,000 tiered and randomized-matrix draws, 944 multi-level one-at-a-time points, twenty Morris trajectories, a 1,024-row Sobol base design on eight leading parameters, and four structural variants beside the base, five architectures in all. In the 400-seed structural comparisons, referral loss produces lower existence, endpoint viability and mean final membership than pure attraction loss in all five architectures. The parameter screens report strict, tied and reversed draws separately and do not turn that finite-domain result into a universal claim.

No parameter is estimated. The contribution is a correspondence, a derivation, an accounting of what the analysis cannot support, and a set of falsifiable predictions, several testable with records AA service structures already keep.

Scope and ethics. This paper does not reproduce the text of the Twelve Steps or Twelve Traditions, which is copyrighted by Alcoholics Anonymous World Services, Inc.; short paraphrases are used throughout. AA is not affiliated with this work and, by its Sixth Tradition, could not be. Nothing in this paper can assess any individual's recovery, and it must not be used to do so. AA's own publications are read where they can be lawfully obtained and are never reproduced at length; a lawful copy, where the project holds one, is a git-ignored local file and is never committed. The historical material is drawn from independent scholarship, from those readings, and from primary temperance sources in the public domain.

Contents

1 Introduction **4**
- 1.1 Roadmap . . . 4

2 Literature Review **5**
- 2.1 Effectiveness and mechanisms of AA . . . 5
- 2.2 Formal models of drinking and recovery . . . 5
- 2.3 Skill formation . . . 5
- 2.4 Social learning and the wisdom of groups . . . 5
- 2.5 Economics of religion, clubs, and teams . . . 6
- 2.6 Nineteenth-century mutual-aid temperance . . . 6
- 2.7 The gap . . . 6

3 Governance as an Aggregation Mechanism **6**
- 3.1 Framework . . . 6
- 3.2 The mapping . . . 7
- 3.3 Failure modes, quantified . . . 7
- 3.4 Obstruction is a property of scaling, not of severity . . . 8
- 3.5 Rotation must scale . . . 9
- 3.6 Rotation may be aimed at the wrong positions . . . 9
- 3.7 Touring speakers: a closed form . . . 10

4 Deriving the Step–Tradition Coupling **10**
- 4.1 A natural conjecture, tested . . . 10
- 4.2 Method . . . 11
- 4.3 Results . . . 11
- 4.4 Six robustness designs, and what each can and cannot see . . . 12
- 4.5 What the coupling results rest on . . . 14

5 The Steps as a Multistage Technology **15**
- 5.1 Identification, anchoring, and design . . . 15

6 Membership Dynamics Under the Open Door **16**
- 6.1 The institutional constraint, restated . . . 16
- 6.2 Model summary . . . 16
- 6.3 Three channels of decline . . . 17
- 6.4 The gap between decline and death . . . 18
- 6.5 Comparative statics under variance control . . . 18
- 6.6 Sensitivity, outcome definitions, and scope . . . 19

6.7 Founding composition: an unresolved comparison 20

7 The Comparative Case **21**
7.1 The received account, and what the primary source says 21
7.2 Prominence without anonymity 21
7.3 Maxwell, and a priority problem stated plainly 22
7.4 What the case cannot do 22

8 Limitations **22**
8.1 The largest one, stated first 22
8.2 The remaining limitations 23

9 Falsifiable Predictions **25**

10 Conclusion **25**

References **26**

A The Whole Paper in Plain Language **30**
A.1 What this paper is about 30
A.2 The big finding: why "nobody's in charge" actually works 30
A.3 What it looks like when the rule breaks 31
A.4 Something the Traditions forgot to mention 31
A.5 The steps don't match the traditions by number 31
A.6 Can you skip a step? Nobody has ever checked 32
A.7 What can actually go wrong for a group 32
A.8 Does it matter who starts a group? 33
A.9 The movement that came before, and what it actually did 33
A.10 How much of this should you believe? 34
A.11 The one-sentence version 34

B Reproducibility Note **35**

1. Introduction

Two literatures have circled Alcoholics Anonymous without meeting. The clinical literature asks whether AA works and through what mechanism; a Cochrane review finds, with high certainty, that manualized twelve-step interventions raise continuous abstinence at twelve months over established alternatives (Kelly, Humphreys, and Ferri 2020), while the mechanism literature is usually read as crediting changes in participants' social networks (Section 2.1). The formal-modeling literature has produced compartmental models of drinking dynamics (Sánchez et al. 2007; Sharma and Samanta 2015), an agent-based model of drinking as social contagion (Gorman et al. 2006), and individual-level dynamical models of behavior change fitted to clinical data (Banks et al. 2014, 2017). Neither literature has modeled the content of the Twelve Steps as a structured process, and neither has modeled the Twelve Traditions at all.

The omission is notable because the Traditions constitute an unusually clean object of institutional study. They are a written constitution for a radically decentralized organization: codified in 1946 from a decade of documented group failures, formally adopted in 1950, and unamended in substance since. They govern through autonomous groups with no enforcement mechanism, no hierarchy, and no budget above the group level. Organizations that hold a governance rule fixed for three-quarters of a century while scaling through six orders of magnitude are rare, and rarer still is one whose founding documents record the failure modes the rules were written to prevent.

The central claim is narrow, formal, and interpretive at one joint. Three of the Traditions (group conscience with servant leadership, Tradition 2; the refusal to organize hierarchically, Tradition 9; and anonymity with principles placed before personalities, Tradition 12) share an effect that is invisible when they are read as ethics and immediate when they are read as a constraint on a stochastic matrix: they prevent any member from acquiring a non-vanishing share of the group's aggregate attention. That is precisely the condition under which DeGroot (1974) belief averaging is asymptotically wise (Golub and Jackson 2010). To our knowledge, this correspondence has not previously been stated.

The joint is worth naming before anything is built on it. The theorem is proved and was read at source. The mapping from three sentences of a fellowship's constitution onto the theorem's hypothesis is a reading of those sentences, arrived at by the author, and *no computation anywhere in this paper touches it.* Everything downstream inherits that status. Section 8.1 states what would settle it and Section 9 states what evidence would overturn it.

Three further contributions support the central one. We derive the coupling between Steps and Traditions from first principles, testing and rejecting the natural conjecture that the two lists pair by index, a reading the historical record already casts doubt on. We reformulate the Steps as a multistage production technology in the form of Cunha, Heckman, and Schennach (2010), under which the informal rule that steps cannot be skipped becomes the Leontief limit of a CES family and its strictness a single estimable parameter. And we build a stochastic membership model disciplined by an institutional constraint central to AA's design: under Tradition 3, an AA group cannot refuse *membership*, so "closing the door" in the sense of excluding a person from the fellowship is not an action available to any group, and open-door adherence operates through newcomer retention rather than newcomer arrival. The constraint proves informative, separating empirically distinguishable channels of group decline.

1.1 Roadmap

Section 2 reviews the relevant literatures. Section 3 states the aggregation framework and the mapping from Traditions to its conditions, with the rotation-scaling and obstruction-scaling corollaries and the touring-speaker structure. Section 4 derives the Step–Tradition coupling and reports six robustness designs against it. Section 5 presents the CES formulation of the Steps and its identification strategy. Section 6 describes the simulation model, its results, and

the sensitivity analysis that bounds them. Section 7 gives the comparative historical case. Section 8 collects limitations; Section 9 states predictions, marking where the analysis above already bears on them; Section 10 concludes. Appendix A restates the paper in plain language. Appendix B is a reproducibility note.

2. Literature Review

2.1 Effectiveness and mechanisms of AA

The Cochrane review of Alcoholics Anonymous and twelve-step facilitation (Kelly, Humphreys, and Ferri 2020) finds, with high certainty, that manualized interventions raise continuous abstinence at twelve months over established alternatives such as cognitive behavioral therapy (two trials, 1,936 participants), and that they do about as well on most other drinking outcomes. Most of the mechanism literature credits changes in a participant's social network: Kaskutas, Bond, and Humphreys (2002) find that AA's effect on drinking outcomes is mediated substantially by changes in the network's composition. The clearest dissent is Rynes and Tonigan (2012), whose fully lagged prospective design finds that the benefit of having a sponsor is not carried by change in the abstinent social network. Measurement instruments for affiliation and involvement are established (Humphreys, Kaskutas, and Weisner 1998; Tonigan, Connors, and Miller 1996; Greenfield and Tonigan 2013). The helper-therapy principle (Riessman 1965) and its AA-specific test (Pagano et al. 2004) address whether helping others benefits the helper. Galanter (1981) supplies an earlier account of why large-group affiliation relieves distress, which is the closest antecedent to this paper's treatment of the group as a producer of resources members consume.

2.2 Formal models of drinking and recovery

Compartmental epidemic-style models treat drinking as transmissible (Sánchez et al. 2007; Sharma and Samanta 2015). Agent-based work has modeled drinking as social contagion among agents moving between locations, one of which is a bar that concentrates drinkers (Gorman et al. 2006). Individual-level dynamical models fitted to clinical data are due to Banks and coauthors (2014, 2017). The relapse literature supplies the nonlinear, multiple-equilibrium structure this paper's individual model borrows (Hufford et al. 2003; Witkiewitz and Marlatt 2004, 2007), and a double-well potential model of relapse supplies the strongest empirical warrant for a two-well landscape (Fatimah, Hunter, and Bornovalova 2025), though Sections 6.2 and 8.2 report that the corrected model is rarely bistable in its own endpoint environment.

2.3 Skill formation

The Steps are treated here as a multistage technology in the sense of Cunha and Heckman (2007) and Cunha, Heckman, and Schennach (2010), with self-productivity, dynamic complementarity, and a CES aggregator whose substitution parameter carries the substantive content. Ben-Porath (1967) is the antecedent. Identification of latent skill from noisy proxies follows Schennach (2004) and Hu and Schennach (2008).

2.4 Social learning and the wisdom of groups

DeGroot (1974) gives the averaging model. Golub and Jackson (2010) give the condition under which such averaging is asymptotically wise, together with the three obstructions to it: prominent agents receiving non-vanishing attention, imbalance between attention given and received, and insufficient dispersion between subgroups. This paper's central section is an application of that result and nothing more; the theorem is not extended.

2.5 Economics of religion, clubs, and teams

Iannaccone (1992) explains costly requirements as screening devices against free-riding in collectives; Lembke (n.d.) applies the frame to AA directly. Holmström (1982) gives the team production problem that a group producing a non-excludable good faces. Ostrom (1990) supplies the design principles for self-governing common-pool institutions, of which the Traditions are an unusually pure instance. Angrist (2014) and Carrell, Sacerdote, and West (2013) supply the warning this paper takes most seriously about interventions built on measured peer effects; see Section 6.7.

2.6 Nineteenth-century mutual-aid temperance

The comparative case in Section 7 rests on primary temperance sources read at source rather than on the secondary literature that descends from AA's own account: Grosh (1842), the Washingtonian movement's own pocket manual; the autobiographies and histories of Hawkins (1862), Marsh (1866), Gough (1869), Eddy (1887), and Blair (1888); and the later scholarly treatments of Fehlandt (1904), Crothers (1911), and Krout (1925). Maxwell (1950) is the one sociological comparison of the two fellowships we have found. Kurtz (1991) is the standard scholarly history of AA and is used for the Traditions' drafting history.

2.7 The gap

No prior work, so far as we have found, treats the Traditions as a formal constraint on an influence structure, derives the Step–Tradition coupling rather than asserting it, or models group-level membership dynamics under the institutional constraint that a group cannot refuse membership. The gap is the paper's occasion; whether the paper fills it well is the subject of Section 8.

3. Governance as an Aggregation Mechanism

3.1 Framework

A group of N members holds beliefs about a matter of collective business. Let A be the row-stochastic trust matrix, A_{ij} the weight member i places on member j, and let beliefs update by $b(t+1) = A\,b(t)$. Under strong connectivity and aperiodicity, beliefs converge to a consensus equal to $s^{\top} b(0)$, where s, the influence vector, is the normalized left stationary vector of A. Rows describe whose beliefs a member uses; columns describe direct attention received. Neither row equality nor column sums alone are long-run influence. Under the regularity conditions in Golub and Jackson (2010), aggregation requires the largest stationary weight to vanish: $\max_j s_j \to 0$.

The error is available in closed form, which is how Table 3 is computed rather than simulated. Assume here that the initial errors are iid Gaussian with standard deviation σ. Then the consensus is $\mu + \sum_j s_j e_j$, a normal variable with mean zero and standard deviation $\sigma\|s\|$, and for a mean-zero normal the expected absolute value is its standard deviation times $\sqrt{2/\pi}$. So

$$\mathbb{E}\,|\text{consensus} - \mu| \;=\; \sigma\,\|s\|\,\sqrt{2/\pi}. \tag{1}$$

Under equal weighting $s_j = 1/N$, so $\|s\| = N^{-1/2}$ and the error is exactly $\sigma\sqrt{2/\pi}\,/\sqrt{N}$. The single-member baseline is the same expression at $N = 1$: $\sigma\sqrt{2/\pi} = 0.798$ at $\sigma = 1$. Every figure in this section is deterministic algebra under that Gaussian benchmark, not Monte Carlo, and carries no sampling error. Independence and a common variance without the Gaussian premise are not sufficient for Equation 1.

Tradition (paraphrase)	Formal content	Role
2. Group conscience; leaders serve, do not govern	Offices are hypothesized not to confer persistent attention received	Proposed mechanism reducing stationary concentration
9. No hierarchy; service rotates	Office-linked attention is periodically reassigned across a broad pool	Proposed mechanism preventing persistent stationary concentration
12. Anonymity; principles before personalities	Some status cues on which attention may condition are suppressed	Proposed mechanism reducing one source of concentration
1. Common welfare first	A strongly connected	Precondition for convergence
4. Group autonomy	No cross-group influence aggregation	No prominent group at higher levels
3. Desire to stop drinking is the only requirement for membership	N unbounded; no screening on membership	Makes the asymptotic regime the relevant one

Table 1: The mapping. **This table is an interpretation of the Traditions' wording, not a result.** It is the paper's central and least verified step.

3.2 The mapping

Proposition 1 (Vanishing stationary influence) *For a sequence of row-stochastic influence matrices satisfying the convergence and signal conditions in Golub and Jackson (2010), beliefs aggregate if the largest normalized stationary influence vanishes,* $\max_j s_j \to 0$. *Persistent stationary concentration on a bounded set is sufficient for failure. A doubly stochastic matrix is a special case: its stationary vector is uniform and* $s_j = 1/N$.

The institutional claim is a separate hypothesis: Traditions 2, 9, and 12 may make persistent stationary concentration less likely. Their wording does not by itself imply exchangeability or double stochasticity. Conversely, violating a Tradition is not sufficient for aggregation failure; names, offices, and hierarchy can exist while $\max_j s_j$ still vanishes.

Remark 1 (The status of Proposition 1) *The established mathematics and the proposed mapping must not be merged. Reading "leaders are trusted servants; they do not govern" as a mechanism limiting attention received is an interpretation of a sentence, and a reader may reasonably hold that Tradition 2 is about humility rather than weighting, or that anonymity is chiefly protective of individuals and only incidentally structural. Nothing computed anywhere in this paper validates that reading. Exchangeability is an additional assumption, not a consequence of the absence of a named office. The testable institutional hypothesis is directional: the Traditions reduce persistent concentration in* s, *not that they make every realized weight equal.*

3.3 Failure modes, quantified

Tables 2 and 3 report the two quantities separately. They are distinct and are easily conflated: at $N = 10$ the flat regime's maximum influence weight is 0.100 and its consensus error is 0.252, and only the first is bounded by construction.

Under flat weighting, error declines as exactly $N^{-1/2}$; under a dominant member or an entrenched caucus it plateaus. The analytic limits confirm the tables: dominant tends to $0.35\sqrt{2/\pi} = 0.279$; the caucus tends to 0.230; the closed core sits at exactly $\sqrt{2/\pi}/\sqrt{5} = 0.357$, the error of a five-member group, because the influence of every member outside the core is identically zero at every N, which is why that column does not vary with N at all.

The closed core is qualitatively worse than the other two and we did not expect that before computing it. The first two put a floor under the group's error. The third deletes the rest of

N	Flat	Dominant	Caucus of 3	Closed core of 5	Rotating, pool 12
10	0.100	0.350	0.167	0.200	n/a
50	0.020	0.350	0.167	0.200	0.041
250	0.004	0.350	0.167	0.200	0.032
500	0.002	0.350	0.167	0.200	0.030

Table 2: Maximum influence weight $\max_j s_j$. Constructions: *dominant*, one member receives 0.35 of every row, remainder split evenly; *caucus*, three members receive 0.50 of every row between them; *closed core*, five members receive 0.45 of every row and distribute their own attention only among themselves; *rotating*, one of $R = 12$ members holds share 0.35 in each term, time-averaged over the cycle (undefined at $N = 10$, where the pool exceeds the group). Exact to the digits shown; no sampling error.

N	Flat	Dominant	Caucus of 3	Closed core of 5	Rotating, pool 12
10	0.252	0.328	0.275	0.357	n/a
50	0.113	0.289	0.238	0.357	0.132
250	0.050	0.281	0.232	0.357	0.093
500	0.036	0.280	0.231	0.357	0.087

Table 3: Mean |consensus − truth| at $\sigma = 1$, from equation (1), against a single-member baseline of 0.798. Same constructions as Table 2.

the group from the calculation: in the limit all influence accrues to the closed subgroup and the group's accuracy is exactly that of the subgroup alone, however many other people are present. This is a direct consequence of Golub and Jackson's imbalance condition and standard for absorbing sets in Markov chains; what is worth reporting is the magnitude.

Consensus is reached in every regime. Concentration does not produce visible dysfunction, only degraded accuracy delivered with undiminished confidence. Both of the first two pathologies have vernacular names in AA: the old-timer whose view settles every group conscience, and the caucus that has decided before the business meeting convenes.

Severity is not the point, and it does not take much. The illustration uses $\alpha = 0.35$, which is a great deal, and a reader could take from it that only substantial dominance matters. A member holding five per cent of every row floors the consensus error at 0.0465 against a flat 0.0252 at $N = 1000$, a factor of 1.84, and that factor grows without bound as the group grows because the flat error keeps falling and the other does not. At ten per cent the factor is 3.3.

3.4 Obstruction is a property of scaling, not of severity

Golub and Jackson's three obstructions (prominent agents, imbalance, insufficient dispersion) map naturally onto the dominant old-timer, the member whose sponsorship lineage listens to him but who listens to nobody, and the clique. **That mapping was asserted before it was computed, and computing it changed it.** The three obstructions are properties of *sequences* of societies as they grow, not properties of a room.

Hold the magnitudes fixed and neither the clique nor the imbalanced member obstructs anything. A clique of three giving a tenth of its attention outward holds, in aggregate, 0.750 of the influence at $N = 10$, 0.375 at $N = 50$, 0.107 at $N = 250$ and 0.029 at $N = 1000$. A member receiving twenty times the attention he gives holds 0.690 at $N = 10$, 0.290 at $N = 50$, 0.074 at $N = 250$ and 0.020 at $N = 1000$. Both shares vanish, both satisfy the condition, and neither is an obstruction.

Now let the same two practices scale with the group. A clique whose inwardness approaches one as the group expands holds 0.167 at every size tested from 10 to 1000. A member whose attention advantage grows in proportion to the group runs from 0.182 at $N = 10$ to 0.167 at $N = 1000$, converging to a positive share rather than settling on one exactly. Neither falls toward zero, and that is the whole of what makes them obstructions.

Corollary 2 (Obstruction scaling) *Whether a concentration of attention obstructs group learning is determined by how it scales with N, not by how severe it is at any one N. The diagnostic question about any concentration is therefore not how large it is but whether it would still be there if the group doubled.*

It is easy to read the three obstructions as three things a room can have at a given moment, and on that reading a tight clique is an obstruction wherever it appears. It is not. The general form of the point is the rotation result that follows.

3.5 Rotation must scale

Proposition 3 (Rotation breadth) *If service rotates over a pool of R members, each officeholder attracting attention share α, time-averaged maximum influence is approximately α/R plus residual flat weight. With R fixed as N grows, $\max_j s_j$ is bounded below and the wisdom condition fails despite rotation.*

At $N = 400$ and $\alpha = 0.35$, the sweep over R gives maximum influence 0.118, 0.060, 0.031, 0.016, 0.009, 0.005 and 0.003 at $R = 3, 6, 12, 25, 50, 100$ and 400, against a flat benchmark of 0.0025. The α/R approximation tracks the computed value closely at every point, which is the check that the mechanism is the one claimed. Holding the pool at twelve while the group grows, maximum influence converges to a floor (0.041, 0.035, 0.032, 0.031, 0.030 at $N = 50, 100, 200, 400, 800$) while the flat benchmark does not (0.020, 0.010, 0.005, 0.0025, 0.0013). The ratio between them runs 2.1, 3.5, 6.4, 12.3, 23.9. That divergence is the entire result.

The pool required to come within a factor of two of flat is 13 at $N = 50$, 26 at 100, 52 at 200, 104 at 400 and 208 at 800: **twenty-six per cent of the membership throughout**, which is the operational form of the corollary.

On the size of the effect, against overstating it. A fixed pool of twelve at $N = 400$ should not be described as approaching the caucus regime. At 0.031 maximum influence and 0.089 error it sits between the flat and concentrated cases and is closer to flat on both measures, against a caucus of three at 0.167 and 0.231. It is a real but moderate permanent cost, not a failure of the same kind as a caucus.

3.6 Rotation may be aimed at the wrong positions

The rotation result prices a pool that is too narrow. A separate objection, raised by the fellowship's own commentary, is that the pool may be irrelevant. *Twelve Steps and Twelve Traditions* (AAWS 1953) describes a group's rotating committee as sharply limited in authority, states that in no sense whatever can its members govern or direct the group, and then locates leadership elsewhere: in "elder statesmen," former officeholders who are called the real and permanent leadership, who are said to become the voice of the group conscience, and to whom a perplexed group inevitably turns. Those members hold no office and therefore rotate out of nothing.

We model this directly. Let e members hold share α_e of every row collectively, with no rotation, and let the remaining attention be flat. This is not the dominant-agent family reparameterised: the share is collective and there is no cycle to time-average.

Proposition 4 (Advisory concentration) *Under the structure above, $\max_j s_j \to \alpha_e/e$ as $N \to \infty$, which is bounded away from zero for any $\alpha_e > 0$. The wisdom condition fails independently of the rotation pool R, since the concentration lies outside it.*

With $e = 3$ and $\alpha_e = 0.10$, maximum influence is 0.1233, 0.0513, 0.0369 and 0.0342 at $N = 10, 50, 250$ and 1000, against the predicted floor of 0.0333 and a flat benchmark falling to 0.0010. Expected consensus error is 0.2552, 0.1214, 0.0681 and 0.0525 against flat values of

0.2523, 0.1128, 0.0505 and 0.0252: a penalty of one per cent at $N = 10$ rising to a factor of 2.08 at $N = 1000$.

The decision-relevant contrast is whether the twenty-six per cent corollary rescues such a group. It does not. Fixing $R = \lceil 0.26N \rceil$ at every size and reporting $\max_j s_j$ as a multiple of the flat benchmark: rotation alone holds at 2.0, 2.0, 2.0 at $N = 50, 250, 1000$, which is a self-consistency check since twenty-six per cent was defined as the within-a-factor-of-two pool. Adding three elders at $\alpha_e = 0.10$ gives 2.5, 9.2, 34.2. At $\alpha_e = 0.20$, 4.1, 17.4, 67.4. With five elders at $\alpha_e = 0.20$, 2.7, 10.7, 40.7.

Scope, and what these numbers are not. The parameters are illustrative, not estimated. No source counts how many members a group defers to on difficult questions or measures the attention they receive, so $e = 3$ and $\alpha_e = 0.10$ were chosen to be conservative rather than fitted. The qualitative result is parameter-free, since $\alpha_e/e > 0$ for any positive α_e; every specific multiple is not. Note also that this is a claim about a practice AA's commentary describes and endorses, not about the Traditions, which continue to satisfy the criterion in Proposition 1. The two are consistent, and the distinction is the paper's recurring one between what a rule specifies and what a fellowship does.

3.7 Touring speakers: a closed form

A movement's attention structure can be modeled as bipartite: flat attention within local societies, plus cross-society attention flowing through touring speakers. Let each member give fraction *out* of their attention to the speakers and let the speakers return fraction *back* of theirs to the general membership. Then

$$\text{speakers' share of total influence} \;=\; \frac{\text{out}}{\text{out} + \text{back}}, \tag{2}$$

exact to twelve decimal places, **independent of the number of members and of the number of speakers**. Five speakers receiving three tenths of the movement's attention and returning two tenths of their own hold six tenths of the influence in a movement of ten and in a movement of a thousand alike.

This is stronger than the argument it replaces. It is not that the influence vector converges on the speakers as the movement grows; the speakers' share is fixed by a ratio and does not move with size at all, because growth adds members to the denominator of the local channel and to the numerator of the speaker channel in equal measure. Section 7 applies it.

A degenerate case, and why the parameter *back* is not decoration. With back $= 0$ the speakers form a closed communicating class, the chain is not strongly connected, Golub and Jackson's theorem does not apply, and the speakers hold *all* the influence at every size and every level of *out*, including 0.1. That is a true statement about an absorbing set and a useless one about a fellowship. The first version of this construction had that defect and reporting it would have been a vacuous-robustness error.

4. Deriving the Step–Tradition Coupling

4.1 A natural conjecture, tested

Two ordered lists of twelve, printed in a single volume, invite the conjecture that Step i draws on Tradition i. The conjecture merits testing rather than adoption. The historical record gives grounds for doubt: the Traditions were codified from 1946 essays written roughly a decade after the Steps, in response to specific organizational crises, and no AA doctrine pairs the lists by index.

That last clause was previously an assertion about literature the project had not read. It has since been checked. *Twelve Steps and Twelve Traditions* (AAWS 1953) is the only work treating both lists at length, one chapter each, by the same author, and is therefore where an

intended pairing would surface. Across its twelve Tradition chapters, none cites the Step of its own number; across its twelve Step chapters, the word *Tradition* does not occur at all. The single indexed cross-reference in the volume runs off-index: the chapter on Tradition 8 mentions the Twelfth Step, to distinguish paid service work from twelfth-step work itself. The count is reproducible in minutes from the files AAWS publishes free, and the procedure is stated here rather than only its result: in each Tradition chapter, count references to the same-numbered Step; in each Step chapter, count occurrences of *Tradition*. The stakes are concrete. Under index-pairing, Step 5 (telling one's inventory to another person) would depend on Tradition 5 (primary purpose), though what the step plainly requires is confidentiality, which is Tradition 12; and Step 12 (carrying the message) would couple to anonymity, though Tradition 5 states nearly the same sentence. We therefore derive the coupling from first principles and test the conjecture against the result.

4.2 Method

To avoid re-deriving the same mistake, no direct step-to-tradition mapping is permitted. An intermediate layer of eight group-produced resources is introduced (admission and standing, identification, living proof, confidential audience, counsel, a recipient for twelfth-step work, continuity, and normative pressure), and the coupling is computed as

$$B \;=\; S\,G^{\top}, \tag{3}$$

where $S[i, r]$ records how much Step i's execution consumes resource r (written by asking what each step requires) and $G[j, r]$ records how much Tradition j governs the supply of r (written independently of S). Both matrices are hand-written judgments. That is the whole of their evidentiary status, and Section 4.5 does not soften it.

4.3 Results

Table 4 gives the unperturbed coupling.

Step	Principal	Value	Runner-up	Value	Index-mate	Value	Its rank
1 admit	T3	1.22	T1	0.99	T1	0.99	2
2 believe	T11	0.82	T5	0.72	T2	0.13	7
3 decide	T2	0.31	T1	0.27	T3	0.01	7
4 inventory	T1/T2 tie	0.35	—	—	T4	0.00	8
5 tell someone	T12	1.08	T1	0.65	T5	0.12	5
6 willing	T1	0.25	T2	0.24	T6	0.00	7
7 ask	T1	0.17	T2	0.13	T7	0.00	7
8 list harms	T2	0.43	T1	0.29	T8	0.07	4
9 amends	T2	0.90	T1	0.48	T9	0.00	8
10 daily	T1	0.94	T2	0.43	T10	0.00	8
11 connect	T1	0.47	T2	0.21	T11	0.11	4
12 carry it	T5	1.25	T3	1.10	T12	0.12	6

Table 4: The author-coded semantic overlap $B = SG^{\top}$, unperturbed. It is not the executable state-update map. Competition ranks use 1+ the count of strictly larger entries, so exact ties share rank. Not one Step has its index-mate in the maximizing set. Load per Tradition, exact: T1 6.52, T2 3.89, T5 3.88, T3 2.69, T11 2.69, T12 2.62, T8 1.11, and exactly zero for T4, T6, T7, T9 and T10.

1. **The author coding contains a two-tier division.** Five Traditions (autonomy, non-endorsement, self-support, non-hierarchy, and no outside issues) have identically zero rows in G. Those zeros were assigned by the author; the matrix multiplication reveals their consequences but does not independently discover or validate the classification.

2. **Unity has the largest raw semantic load under this coding.** Under index-pairing, unity fed one step; in the authored matrices it overlaps with resources consumed by nearly every Step. The multiplication summarizes those judgments; it does not supply independent evidence for them.
3. **The index-pairing conjecture is rejected on all twelve counts.** Step 5's principal supplier is Tradition 12 and Step 12's is Tradition 5, the two correspondences a reader would guess, reversed.
4. **Classification is not always intuitive.** Tradition 8 (non-professionalism), which a functional reading might class as protective, lands in the enabling tier with small load, because it governs the identity of the confidential hearer, a fellow member rather than a clinician.

4.4 Six robustness designs, and what each can and cannot see

The trivial-count decomposition, stated first because it deflates the headline. T4, T6, T7, T9 and T10 have identically zero rows in G, so Steps 4, 6, 7, 9 and 10 have index-mate entries of exactly zero, and index-pairing cannot hold for them under *any* perturbation that preserves sparsity. Both multiplicative designs below preserve sparsity. The consequence is checkable and was checked: at every level, the proportion of draws in which index-pairing fails on all twelve equals the proportion in which it fails on the seven non-trivial Steps, to the last draw (98.75 and 98.75 at ±15 per cent; 85.50 and 85.50 at ±30; 72.45 and 72.45 at ±50; 65.15 and 65.15 at ±75; 40.60 and 40.60 structurally). **Five of the twelve counts are not evidence, and no multiplicative design could have made them so.**

Design 1, multiplicative jitter. Every entry of S and G is multiplied by an independent uniform draw on $[1-L, 1+L]$ for $L \in \{0.15, 0.30, 0.50, 0.75\}$; 2,000 draws per level, seed 3. This represents a reader who disagrees with the magnitudes by up to L and agrees about which cells are empty.

Design 2, structural randomization. Every non-zero entry of S and G is replaced by an independent uniform draw on $[0.05, 1.00]$; zeros stay zero. 2,000 draws, seed 23. This represents a reader who accepts only the pattern of which Tradition touches which resource and rejects every magnitude we chose. A parallel 1,000-draw screen passes randomized nonzero matrices through the full executable model: the referral-versus-pure-attraction comparison is strict in all 1,000 draws on final membership and endpoint viability and in 999, with one tie, on existence. Full-adherence viability in all three screening seeds holds in only 784 draws. The comparison and the absolute outcome therefore cannot borrow one another's robustness.

Design	T1 leads	Index-pairing wrong, all	Step 5 → T12	Step 12 → T5
jitter ±15%	100.0 [99.8, 100.0]	98.8 [98.2, 99.2]	100.0 [99.8, 100.0]	89.7 [88.3, 91.0]
jitter ±30%	100.0 [99.8, 100.0]	85.5 [83.9, 87.0]	99.5 [99.1, 99.7]	67.3 [65.3, 69.4]
jitter ±50%	98.0 [97.3, 98.5]	72.5 [70.5, 74.4]	88.7 [87.2, 90.0]	52.9 [50.7, 55.1]
jitter ±75%	86.8 [85.3, 88.3]	65.2 [63.0, 67.2]	71.5 [69.5, 73.4]	43.8 [41.6, 45.9]
structural	**75.4 [73.5, 77.2]**	**40.6 [38.5, 42.8]**	**17.5 [15.9, 19.2]**	**27.6 [25.7, 29.7]**

Table 5: Percentage of draws in which each claim holds. Wilson intervals at 95 per cent on $n = 2{,}000$. The three 100.0 entries are 2,000 of 2,000 and should be read as "not observed to fail", not as certainty.

Step 12 is the fragile row and the reason is the margin, not the level: its top two are T5 at 1.25 and T3 at 1.10, and both Traditions govern the recipient resource (T5 at 0.9, T3 at 0.8), so a perturbation moving them in opposite directions flips the winner.

Design 3, the threshold test, which is the only one that can see the five. For Step i, let w_i be the row sum of S (total consumption across the eight resources) and b_i the largest entry in row i of B excluding the index-mate's own entry. If Tradition i governed every resource at a uniform strength c, its entry in row i would be $c\,w_i$, so index-pairing holds at Step i exactly when

$$c > c_i^* = b_i / w_i. \tag{4}$$

This is one division per Step, carries no sampling error, and is the only design here that can turn a structural zero into a non-zero.

Step	Index-mate	Beats	Row sum of S	c^*	c^* / mean live entry
1	T1	1.22	2.40	0.508	1.36
2	T2	0.82	1.70	0.482	1.29
3	T3	0.31	0.70	0.443	1.18
4	**T4**	0.35	0.80	**0.438**	1.17
5	T5	1.08	1.70	0.635	1.70
6	**T6**	0.25	0.60	**0.417**	1.11
7	**T7**	0.17	0.40	**0.425**	1.14
8	T8	0.43	0.80	0.538	1.44
9	**T9**	0.90	1.50	**0.600**	1.60
10	**T10**	0.94	1.50	**0.627**	1.67
11	T11	0.47	0.80	0.588	1.57
12	T12	1.25	2.40	0.521	1.39

Table 6: The threshold test. Bold rows are the five protective Traditions, whose index-mate entries are structurally zero and therefore invisible to Table 5. Exact; no sampling error.

The governance matrix has 35 non-zero entries of 96 cells, with mean 0.374, median 0.300, minimum 0.10 and maximum 1.00. *Every* c^* exceeds both the mean and the median. The protective range, 0.417 to 0.627, is not contained in the enabling range, 0.443 to 0.635; it extends below it at both ends. Mean c^* is 0.501 for the five protective Steps, 0.531 for the seven enabling ones, and 0.518 across all twelve. The protective Steps are therefore marginally *closer* to index-pairing holding, by 0.03 on the mean threshold, which is the direction an objector would predict and an order of magnitude smaller than the distance to the mean live entry. Setting c to the mean live entry of 0.374 for every Step simultaneously, all twelve index-mates still lose, by margins from 0.02 at Step 7 to 0.44 at Step 5.

What the test licenses is the statement that index-pairing would require the index-mate to govern its own Step's needs more strongly than a typical entry in the matrix, uniformly across all twelve. What it does not license is any statement about whether the zeros are correctly *placed*: it holds the sparsity pattern's origin fixed and prices only its consequences. It is a sensitivity analysis of a judgment, not a test of it.

Design 4, the resource-list test. Delete each of the eight resources in turn, then merge each pair, then delete each pair: sixty-four resource lists in all. Only Step 1 and Step 2 ever regain their index-mates, in fifteen variants and one respectively, and every one of the fifteen Step 1 failures involves removing or merging the admission resource, which is precisely what gives the open door its lead there. This establishes that the list is no *finer* than it needs to be. It cannot establish that the list is fine *enough*, because inventing a ninth resource requires a judgment about what it contains and cannot be done by rearranging the eight.

Design 5, the reassignment test. Kurtz (1991) records that in some later AA literature the concept conveyed by *single-purposed* was obfuscated by substituting *unity*. If the governance matrix absorbed that semantic drift, unity's primacy is an artifact of it. For each of the eight resources in turn, set $G[\mathrm{T5}, r]$ to the maximum of its current value and $G[\mathrm{T1}, r]$, set $G[\mathrm{T1}, r]$

to zero, and recompute: this transfers unity's entire governance of one resource to singleness of purpose, which is the strongest form of the objection, and it is exact. **Six of the eight transfers leave unity leading**, by margins from 1.44 to 2.54. Transferring continuity gives T5 the lead 5.14 to 4.63; transferring pressure gives it 5.12 to 4.66. So unity's primacy is conditional on two assignments and on nothing else in the matrix. Notably, the single resource-list variant that breaks it (Design 4) drops continuity and pressure together, which is exactly the pair this test identifies: two independent designs agreeing on which two resources carry a result is worth more than either alone. The passage Kurtz names as decisive is in an AA-copyright work that has not been obtained, so the source identified as settling the question is unread. The 1953 commentary, read in full on 10 August 2026, keeps unity and singleness of purpose apart across its chapters on Traditions 1 and 5, which corroborates the assignment without being the passage Kurtz cited.

Design 6, sparsity perturbation. Every design above varies magnitudes and holds the sparsity pattern fixed; a second reader would disagree about the pattern. Flipping cells at random within the seven enabling rows (56 cells, so the two-tier split is held fixed), 2,000 draws per flip count. A flipped cell takes the value 0.5, so a Step's strongest Traditions often tie exactly. A claim counts as holding in a draw only where it holds strictly, since a tie is not support, and the products are computed exactly, every weight being a multiple of 0.01, so the table does not depend on how a machine rounds:

Cells flipped	Index-pairing wrong, all	T1 leads	Step 5 → T12	Step 12 → T5
1	96.4 [95.4, 97.1]	100.0 [99.8, 100.0]	98.7 [98.1, 99.1]	94.9 [93.8, 95.7]
2	92.7 [91.4, 93.7]	99.6 [99.1, 99.8]	96.4 [95.4, 97.1]	90.5 [89.1, 91.7]
4	86.3 [84.7, 87.7]	96.4 [95.4, 97.1]	93.4 [92.2, 94.4]	80.4 [78.6, 82.0]
8	74.8 [72.8, 76.6]	83.6 [81.9, 85.1]	85.4 [83.8, 86.9]	72.0 [70.0, 73.9]
16	51.7 [49.5, 53.9]	50.8 [48.6, 52.9]	70.8 [68.8, 72.8]	52.9 [50.7, 55.0]

Table 7: Sparsity perturbation. Percentage of draws in which each claim holds strictly, with Wilson intervals at 95 per cent on n = 2,000; the 100.0 entry is 2,000 of 2,000. Section 4 tolerates a reader differing on about four of fifty-six enabling cells and does not tolerate one differing on sixteen.

This is a bound, not a measurement, and the distinction is the caveat. A random flip is not a plausible reader. Somebody who thinks self-support governs continuity changes that cell for a reason, and their remaining cells correlate with the reason. Random flips are harsher in respecting no reason and gentler in not concentrating on the cells that carry the results. The sweep answers "how much disagreement, counted in cells" and says nothing about which cells a reader would choose. It also holds the two-tier split fixed, because the split is what defines which rows are available to flip, so the more serious disagreement, a reader who fills a protective row, cannot be represented by any perturbation design at all. Only completed elicitation forms can answer either question, and none has been returned; see Section 8.

4.5 What the coupling results rest on

Two classes of claim in this paper rest on different things and must not be defended the same way.

The executable model's **referral-versus-pure-attraction comparison** survives replacing every nonzero matrix magnitude with a random value while preserving sparsity: 1,000 of 1,000 strict draws on final membership and endpoint viability, and 999 strict plus one tie on existence. The fully adherent group's endpoint viability is not invariant. This result concerns the executable model, not the raw coupling statistics in this section.

The **coupling** claims of this section do not. Index-pairing fails on all twelve in only 40.6 per cent of structurally randomized draws, which is less often than it holds; the Step 5 inversion

survives in 17.5 per cent and the Step 12 inversion in 27.6. These claims rest on the magnitudes in two matrices we wrote, and they must be argued for rather than certified by a robustness percentage earned elsewhere. The one exception is unity's primacy, which survives structural randomization in 75.4 per cent of draws, the highest figure of any claim in this section: three-quarters of the time a random matrix with the same sparsity pattern puts unity first anyway.

A reader who thinks the two matrices are arbitrary should not be persuaded by Section 4, and we would rather say so than borrow an executable endpoint comparison's robustness for a semantic-coupling result that has not earned it.

5. The Steps as a Multistage Technology

> **This section is a reformulation, not the simulation's update rule, and the two must not be read as the same object.** The CES form below is a way of *stating* the ordering claim so that its strictness becomes one estimable parameter. The dynamic model of Section 6 uses the gated-growth equation given in §6.2, which is not a CES aggregator. Nothing in Section 6 depends on the value of ρ.

Let $x_{i,t}$ denote latent practice of step i. Following Cunha, Heckman, and Schennach (2010):

$$x_{i,t+1} \;=\; A_i\Big[\,\gamma_{i1}x_{i,t}^{\rho_i} + \gamma_{i2}x_{i-1,t}^{\rho_i} + \gamma_{i3}G_{i,t}^{\rho_i} + \gamma_{i4}M_t^{\rho_i}\,\Big]^{1/\rho_i}, \tag{5}$$

with weights summing to one, elasticity of substitution $\sigma_i = 1/(1-\rho_i)$, $G_{i,t}$ the resource bundle from Section 4, and M_t maintenance capacity built from Steps 10 to 12. (Cunha, Heckman, and Schennach write the substitution parameter as ϕ and reserve ρ for an outcome share; ρ here is our notation.) Self-productivity is $\gamma_{i1} > 0$; the cross-partial in prior-stage stock and group input is positive throughout (from $+1.73$ at $\rho = -4$ to $+0.08$ at $\rho = 0.5$), so group support is worth more to a member who has done the preceding work: dynamic complementarity in Cunha and Heckman's sense.

Proposition 5 (The ordering rule as a limit) *As $\rho_i \to -\infty$ the aggregator converges to $\min(\cdot)$, so zero prior-stage stock forces zero output: the informal rule that steps cannot be skipped. The rule holds for all $\rho_i \le 0$ and fails for $\rho_i > 0$, where the group input substitutes for the missing stage (output 0.217 at $\rho = 0.3$ with prior stage at zero).*

The reduction converts a widely held but untested claim of practice into a sharp empirical question: within this CES reformulation, the sign of ρ decides whether the Steps are a chain or a menu. Estimating that sign would require an empirical identification design; the calculation here does not estimate it. In the one setting where an analogous parameter has been estimated, children's skill formation, Cunha, Heckman, and Schennach (2010) find it positive in early childhood and strongly negative later for cognitive skill, and negative at both stages for noncognitive skill, so there is no reason to expect a single sign across all twelve Steps.

5.1 Identification, anchoring, and design

Latent practice would have to be measured with a design that addresses error in the inputs, loadings, scale, and group-input endogeneity. Schennach (2004) and Hu and Schennach (2008) describe identification strategies under explicit assumptions, but this paper does not implement them. Its simulation gives the four regressors and proxy loadings to the estimator without error and adds noise only to the outcome. Repeating that exercise shows the modest point that averaging three rescaled output proxies improves resolution relative to one. It does not recover a latent-variable estimator, justify three instruments per latent per wave, or settle the sign of ρ.

The binding obstacle to estimation is endogeneity of the group input: groups direct attention to struggling members and successful members attract sponsees, so G is not exogenous to member state. CHS address the analogous endogeneity of parental investment; adapting their approach is necessary before any fit to panel data and is not attempted here.

6. Membership Dynamics Under the Open Door

6.1 The institutional constraint, restated

Tradition 3 makes the desire to stop drinking the sole requirement for **membership**. It is tempting to render this as "an AA group cannot refuse admission," and that overstates it. AA groups hold closed meetings as a matter of routine, and a closed meeting restricts who is present in a room; it does not remove anyone's membership in the fellowship. The constraint is on membership, not on attendance, and the model is built to that constraint. The distinction was supplied by a reader who knows the rooms, and it is the single correction this work has received that changed the most for the fewest words.

The distinction has modeling consequences. Arrival at a meeting does not depend on the group's welcome in the default model. Tradition 3 instead has two separately controllable paths: its row governs four resource columns, and it adds dropout friction weighted by inverse practice. The second is not an early-tenure variable: the model contains no tenure or cohort state, so a long-tenured low-practice member receives the same weight as a recent arrival at the same practice. Inflow contains an exogenous referral floor independent of the group's attractiveness; exit also includes a practice-independent churn term for relocation and mortality.

Two further design choices follow from the same logic. Carrying capacity is supplied by the established core rather than the population mean, since a room of three veterans and twenty newcomers still contains three people able to carry a low-practice member. And the maintenance-capacity gate, whose role is to model relapse in members with something to maintain, phases in with step index rather than throttling entry-level growth, to which it does not conceptually apply.

6.2 Model summary

Members occupy a twelve-dimensional practice state. Per-step growth, in full, for member m and step i:

$$\frac{dx_i}{dt} \;=\; h(m)\, a_i\, \mathrm{gate}_i\, \mathrm{peer}_i\, C_i^m\, (1 - x_i) \;-\; d_i\, x_i, \tag{6}$$

with

$$\mathrm{gate}_i = x_{i-1}^{p}, \qquad \mathrm{gate}_1 = 1, \tag{7}$$

$$\mathrm{peer}_i = (1 - \beta_i) + \beta_i G_i, \qquad G = S_{\mathrm{norm}} R, \tag{8}$$

$$C_i^m = 1 - w(i)\,(1 - C), \qquad w(i) = 0.05 + (i-1)\tfrac{0.95}{11}, \tag{9}$$

$$d_i = \delta_0 (1 + \psi(1 - x_{i+1})) \;\; (i < 12), \qquad d_{12} = \delta_0. \tag{10}$$

Maintenance capacity C is identical across steps for a given member and is defined in two parts. Own capacity is a Hill gate on M, the mean of that member's Steps 10 to 12, and group support supplies a floor beneath it:

$$C(M) = \frac{M^n}{k^n + M^n}, \quad n = 3.0,\; k = 0.12; \qquad C = C(M) + \big(1 - C(M)\big)\,\omega\,\overline{C}, \quad \omega = 0.75, \tag{11}$$

where $\overline{C}$ is the mean own-capacity across living members, so a member whose own maintenance has collapsed retains a fraction of capacity as long as the group around them has not. Because Steps 10 to 12 sit at the heavy end of the $w(i)$ weighting, maintenance gates its own accumulation.

That architecture can be bistable, but it is not generally so in the corrected model's own endpoint environments: a capability-one high/low-start test separates in 7 of 400 environments and collapses to one low state in the mean environment. Here $w(i)$ is per-step exposure to it, rising linearly from 0.05 at Step 1 to 1.00 at Step 12, which is the formal content of the claim that an arrival has nothing to maintain and a veteran has a great deal; and $h(m)$ is member heterogeneity, a lognormal draw made once when a member arrives and fixed thereafter. It is parameterized as $\exp(N(-\sigma_h^2/2, \sigma_h))$, so its arithmetic expectation is one and changing σ_h changes dispersion without mechanically changing mean capability.

Group resources R are produced from member states through the governance-weighted averages of Section 4, so shortfalls do not compound multiplicatively, with protective Traditions acting solely as multipliers on the enabling ones they guard. Membership is endogenous: arrivals are exogenous referrals plus attraction proportional to members' twelfth-step practice and the attraction path of Tradition 11; exits combine practice-dependent dropout, the Tradition-3 friction path, and churn. Tradition 11 also has a distinct resource-governance path. Integration is Euler at $dt = 0.5$ weeks over a 1,560-week (thirty-year) horizon. This horizon is a reporting choice, not a steady-state assertion. In 200-seed checks, mean membership is 29.34, 21.14, 16.43 and 15.64 at 10, 20, 50 and 100 years. The 100-year condition includes one closure. Paired changes in final membership from $dt = 0.5$ to 1.0, 0.25 and 0.125 weeks are respectively 1.54 $\pm$ 1.78, 0.87 $\pm$ 1.84 and -0.41 $\pm$ 1.74. None resolves at this sample size, so the tested step is adequate for the reported finite-horizon comparisons; the runs do not establish an equilibrium or indefinite persistence.

Parameters: $\delta_0 = 0.06$ per week (unattended half-life 11.6 weeks), $\psi = 0.20$, $p = 1.5$, step top speeds from 0.15 at Step 9 to 0.30 at Step 1. The inventory is 22 continuous scalars, 12 step speeds, 49 non-zero cells in S and 35 in the governance matrix: **118 registered sensitivity values out of 226 listed cells, none fitted.** The model-choice inventory separately records fixed constants, structural zeros, equations, thresholds and experiment-design choices, so 118 is not the count of every authored choice.

Remark 2 (Calibration, and the exact sense in which it fails) *Inflow, dropout and churn were originally set to target a group near forty-five members with an experienced core near nine, roughly a healthy urban meeting. That was calibration to a stylized fact, not a dataset. After the lognormal capability draw was corrected to have arithmetic mean one, the target fails. Across 400 runs the model delivers 17.80 members, 95 per cent half-width 0.88. Among the 394 viable endpoints, the* established *count above 0.1 is 14.13, half-width 0.79, and the* experienced *count above 0.5 is 1.25, half-width 0.20. The two thresholds are not interchangeable. The authored rates are not retuned after observing this failure, and the absolute levels are not estimates of real meetings.*

Remark 3 (Thirty-five parameters that cannot matter at full adherence) *The governance matrix is column-normalized, so governance quality is identically 1 when every Tradition is at 1.0 and the matrix cancels exactly. This is algebra, not simulation. All 35 governance cells therefore produce exactly zero change in every outcome at full adherence, and a sensitivity design that perturbs them and reports no effect has found nothing. Thirty-five of the 118 registered values are in that position.*

6.3 Three channels of decline

Four configurations, 400 paired seeds each, thirty-year horizon, $dt = 0.5$ weeks. "Viable" means more than five members; existence means at least one, and zero is permanent closure. **Quality is established-member practice in viable groups and is conditional throughout; the viable fraction is printed beside it in every row.** Viability carries a 95 per cent Wilson interval; membership and quality carry a 95 per cent half-width from the cross-run standard error.

Condition	Viable y10	N if viable, y10	Quality y10	Viable y30	95% int.	Quality y30
nothing wrong	0.9975 [0.9860, 0.9996]	29.40 ± 1.16	0.3016 ± 0.0052	0.985	0.968–0.993	0.2648 ± 0.0058
invisible	0.9925 [0.9782, 0.9974]	12.98 ± 0.34	0.2949 ± 0.0079	0.985	0.968–0.993	0.2592 ± 0.0071
unreferred	0.660 [0.612, 0.705]	16.47 ± 1.18	0.3475 ± 0.0084	0.0275	0.015–0.049	0.3642 ± 0.0443
unwelcoming	0.905 [0.872, 0.930]	15.70 ± 0.82	0.3597 ± 0.0097	0.5475	0.499–0.596	0.3211 ± 0.0134

Table 8: Decline scenarios, **400 paired seeds**, 1,560-week horizon. *Invisible* sets only Tradition 11's attraction path to zero; *unreferred* sets exogenous inflow to zero; *unwelcoming* sets both Tradition 3 paths to zero. Only eleven unreferred runs are viable at year thirty, so its conditional quality interval is wide.

Membership counted over all runs with closures as zero reaches, at year thirty, 17.80 ± 0.88, 12.38 ± 0.34, 0.51 ± 0.23 and 6.76 ± 0.59 respectively.

Three signatures separate. Attraction loss leaves a smaller remnant sustained by referrals and produces no closure. Referral loss closes 89.5 per cent of groups by year thirty, while the eleven viable endpoints have high conditional practice. Combined Tradition 3 loss closes 25.0 per cent and leaves 54.8 per cent viable.

The mortality profiles follow from structure rather than from hand-tuning. A group on one engine has no floor underneath it: its inflow becomes a function of its own state, which makes the population dynamics multiplicative rather than additive, and a multiplicative process with no floor has an absorbing state at zero. Invisible and unwelcoming groups retain a referral floor, but the latter can still close through its resource and dropout paths. Unreferred groups are the only ones whose inflow itself can go to zero and stay there.

The quality claim, stated exactly. At year ten, established-member practice against a healthy viable group is 0.0067 lower for the invisible condition, 0.0459 higher for the unreferred one, and 0.0581 higher for combined Tradition 3 loss. Membership is lower in all three. Practice can therefore mislead through selection, but the corrected model does not support the stronger claim that the interior of a dying group offers no warning.

6.4 The gap between decline and death

The unreferred condition separates two questions that a single membership series conflates.

Series	y5	y10	y20	y30
unreferred, all runs	21.55 ± 1.01	11.68 ± 1.02	2.50 ± 0.48	0.51 ± 0.23
unreferred, viable only	22.07 ± 1.00	16.47 ± 1.18	11.39 ± 1.37	12.27 ± 3.81
viable fraction	0.970	0.660	0.1725	0.0275 [0.015, 0.049]

Table 9: Unconditional versus viability-conditioned membership under referral loss. Both decline sharply; conditioning still hides the mass of closures but no longer makes membership look stable.

This is why conditioning is load-bearing throughout the paper. In the referral-loss condition both the all-run and viable-only membership series decline, but the all-run series also carries the growing mass of closures. Neither series can stand in for the other.

6.5 Comparative statics under variance control

Discipline preceded comparison: baseline survival is stable across independent seed blocks, and common random numbers reduce the paired standard error of membership comparisons. The reference cross-seed standard deviation is 5.67, while paired standard errors range from 0.22 to 0.33. Each Tradition is degraded singly from 0.85 to 0.5 against a reference group of 13.10 members, over a twenty-year (1,040-week) horizon.

> **The replication budget determines the ranking, and 30 is not enough.** Computed at 30 paired replications, this comparison returns autonomy and self-support at 5.5 members

Tradition	Tier	Members cost	95% half-width	t
T3 mixed adherence	enabling	2.99	0.65	9.1
T11 mixed adherence	enabling	1.88	0.60	6.1
T1 unity	enabling	1.65	0.65	4.9
T12 anonymity	enabling	0.99	0.58	3.3
T5 one purpose	enabling	0.96	0.61	3.1
T4 autonomy	protective	0.88	0.62	2.8
T7 self-support	protective	0.88	0.62	2.8
T2 group conscience	enabling	0.48	0.62	1.5
T6 no endorsement	protective	0.23	0.46	1.0
T10 no outside opinion	protective	0.23	0.46	1.0
T9 no organization	protective	0.02	0.43	0.1
T8 non-professional	enabling	-0.20	0.51	-0.8

Table 10: Single-Tradition degradation, **400 paired replications** under common random numbers. Seven of twelve have intervals excluding zero. T3 and T11 are mixed adherence interventions; their paths are split in the release factorials.

each with $t = 2.6$ as the only pair clearing $|t| > 2.5$, which invites the inference that the protective Traditions lead the ranking, consistent with their derived role as guards on everything else. Under the corrected model, at 400 replications mixed Tradition 3 leads at 2.99 members, mixed Tradition 11 follows at 1.88, and seven comparisons resolve rather than two. **A common-random-numbers design at 30 replications is more efficient than 30 independent runs and is still 30 replications.** Variance reduction buys precision per replication; it does not substitute for replications, and any ranking reported at that budget is an artifact of it.

6.6 Sensitivity, outcome definitions, and scope

The simulation's parameters were subjected to: global multiplicative jitter at three amplitudes over 1,002 draws; a tiered design and a randomized-matrix design at 1,000 draws each; 944 multi-level one-at-a-time perturbations covering all 118 registered values at four distances in each direction; a twenty-trajectory Morris elementary-effects screen over all 118; a Sobol decomposition on a 1,024-row base design over the eight Morris membership leaders; four structural variants that change the model's architecture rather than its numbers; and a resource-list test. Full specifications are in the book's technical appendix, sections A3 to A9.

Which parameters matter. The ordering exponent p has the largest single influence of any parameter on every outcome the multi-level sweep scores. In the three-seed one-at-a-time screen, against a full-adherence baseline of 0.0431 maintenance, moving p alone by 25 per cent in each direction swings maintenance from 0.0027 to 0.2373, a range of 5.45 times baseline; the decay rate δ_0 is second at 3.27 and the step-11 speed third at 2.22. Across the full four-distance ladder the p range is 14.64 times baseline, δ_0 12.98 and member heterogeneity 4.69. The twenty-trajectory Morris screen (five seeds a point, at the 0.85 reference adherence) ranks the same two first on membership μ^*, at 52.44 and 51.21, followed by churn at 33.48, dropout sharpness at 24.30, exogenous referral at 23.46, heterogeneity at 18.90 and the step-6 and step-12 speeds at 18.54 and 15.87. Sobol total-order indices on membership, over those eight leaders at plus or minus 25 per cent, agree: p at 0.576 [0.510, 0.643], δ_0 at 0.373 [0.320, 0.428], dropout sharpness at 0.171, churn at 0.132, heterogeneity at 0.079 and exogenous referral at 0.058, against a noise floor of 0.043 for membership and 0.073 for practice. The two step speeds, at 0.029 and 0.038, sit at or below that floor and are not separated from Monte Carlo error.

The 1,024-row release design gives an admissible first-order column for membership and not for practice. On membership no factor now has $S_1 > S_T$, and the first-order indices sum to 0.693, so the column is admissible: p resolves at 0.404 [0.288, 0.525] and δ_0 at 0.238 [0.156,

0.327], while the remaining six have intervals covering zero and are unresolved rather than zero. On practice the column is still not usable: δ_0 returns $S_1 = 0.421$ against $S_T = 0.417$, violating the identity $S_T \geq S_1$ that holds for any true decomposition, and the practice first-order indices sum to 1.074, whereas a sum of first-order indices cannot exceed one. No practice first-order number is quoted elsewhere. The sums of S_T are 1.456 for membership and 1.464 for practice, and the excess over one is the signature of interaction counted once per factor involved, so interactions are present and are not dominant. The decomposition is conditional on these eight factors and these ranges; it is not a decomposition of the model's total variance.

Structural variants. Four changes to the architecture, not the numbers, are compared with the base model: a flat per-step gate, admission moved to Tradition 3, capacity supplied by all members rather than the established core, and a piecewise-linear rather than hyperbolic member-side capacity response. The last variant still clips at one; its historical filename is shorthand, not a literal absence of saturation. Each of the five architectures is run at 400 paired seeds for five scenarios. Under pure attraction loss, endpoint existence is 1.000, 0.870, 1.000, 1.000 and 1.000; endpoint viability is 0.985, 0.658, 0.985, 1.000 and 0.995; and mean membership is 12.38, 6.61, 12.38, 14.73 and 13.04. Under referral loss the corresponding values are 0.105, 0.120, 0.105, 0.258 and 0.145 for existence; 0.028, 0.055, 0.028, 0.123 and 0.058 for viability; and 0.51, 0.75, 0.51, 2.00 and 1.02 for membership. Thus the referral-loss ordering holds on all three outcomes in all five tested architectures. This is a finite structural audit, not a proof over untested architectures.

> **What the structural audit says, and no more.** In the corrected model, referral loss is worse than pure attraction loss on existence, endpoint viability and mean final membership in the base architecture and all four variants. The earlier result in which the size ordering reversed under three variants came from the retired, uncentred capability model and does not reproduce. The parameter screens remain a separate question and report strict orderings, ties and reversals at each design level. Specific probabilities and memberships remain conditional statements about a constructed model.

What the designs could in principle have found. Two failures are worth naming because both were made in this project. A multiplicative perturbation cannot move a structural zero, so no multiplicative design is evidence about the two-tier split (Section 4.4). And a ±30 per cent jitter cannot tell you whether a result depends on the magnitudes at all, only whether it tolerates small disagreement about them; the structural randomization is the design that answers the first question, and it is the one the coupling claims fail.

6.7 Founding composition: an unresolved comparison

Twenty-five founders with a fixed total practice of 13.75 distributed three ways (even: all at 0.55; concentrated: five at 1.00 and twenty at 0.4375; split: twelve at 0.90 and thirteen at 0.2269), 400 paired seeds each. Membership ends at 17.80, 18.09 and 17.45 with half-widths 0.88, 0.99 and 0.93; endpoint viability is 0.985, 0.9875 and 0.985; established practice is 0.2645, 0.2628 and 0.2660. Relative to even founders, the paired membership differences are 0.29 [-0.90, 1.48] and -0.35 [-1.47, 0.78]. No equivalence margin was prespecified, so the result is unresolved rather than evidence of equality.

The unresolved comparison should be discounted heavily before it is read. Founding practice changes ordering gates, maintenance, resource capacities, dropout and attraction jointly. Resources are computed from aggregates and no member's state appears in another member's growth equation except through those aggregates, so the model has no representation of mentoring, pairing, cliques or sponsorship. **It also has no representation of the mechanism that produced the Carrell–Sacerdote–West result**, which is people choosing whom to associate with inside a group whose composition has been arranged. That experiment engineered

Air Force Academy squadrons from measured peer effects, predicted a gain of 0.053 grade points for the bottom third, and measured a treatment effect of −0.061 on exactly the students it set out to help, because the low-ability cadets re-sorted toward each other. A non-significant contrast from a design without that mechanism is not evidence of practical equality.

The design also cannot separate two things: the conditions differ in the variance of founding practice *and* in the number above the stricter 0.5 experienced threshold (25, 5 and 12). All 25 exceed the 0.1 established threshold in every condition. A resolved difference could still have arisen through any of several state-dependent channels.

7. The Comparative Case

A formal correspondence gains little from a single historical case and can lose a great deal by leaning on one. This section is included because the case bears directly on the paper's central claim, and because the standard account of it, which is what a reader is most likely to bring, turns out to be false.

7.1 The received account, and what the primary source says

The Washingtonian Total Abstinence Society was founded in Baltimore in April 1840 by six working men, in a scene whose earliest surviving account is Harrison (1860) and which reaches most modern readers through Maxwell (1950), spread nationally within four years on claims reaching into the hundreds of thousands, and was effectively finished within a decade. In the literature descending from AA it is standardly described as a movement that died of having no rules, and it is used to illustrate why the Traditions matter. That literature arrives with its moral pre-attached.

The movement's own manual falsifies the account. Grosh (1842), the *Washingtonian Pocket Companion*, printed at Utica and by then in a second edition, carries in its first fifteen pages a definition of principles, directions for organizing and conducting meetings, and a model constitution. Among its provisions: each society independent and subordinate to none; funds controlled by its own members; and nothing sectarian or political admitted to lectures, speeches, singing, or the doings of the society. A footnote records that a Washingtonian mass convention at Utica passed a declaration of principles and a model constitution on 22 February 1842, that it was printed in the *Utica Washingtonian* of 25 February and reprinted in October because of demand, and that a copy should be procured wherever a society is organized. **They had written rules, they had them within two years of founding, and they had a mechanism for transmitting them.**

What they had, in writing, were analogues of four of the Traditions this analysis classifies as protective. So far as the ledger shows, what they had none of were the enabling Traditions: no analogue of Tradition 2, and the opposite of Tradition 12. And on anonymity they took the opposite position deliberately and with an argument: Grosh's directions for a first meeting have joiners rise and call out their names for the secretary, because "publicity and freedom are preferable to private solicitations, whisperings, and secresy in giving the names."

The comparison the case actually supports is therefore not rules against no rules. It is one written code against another, and the provisions missing from the first are the ones the aggregation condition points at.

7.2 Prominence without anonymity

The touring-speaker structure of §3.7 was built for this case. Washingtonian expansion from 1841 to 1843 proceeded through touring speakers addressing local societies, which is exactly a rise in the proportion of total attention carried by a one-directional cross-society channel. By equation (2) the speakers' share of influence is fixed by the ratio of outward to returned

attention and does not fall as the movement grows, so the movement could not have outgrown the exposure.

The exposure was realized. John B. Gough, the movement's most prominent speaker, relapsed publicly in September 1845; opponents seized on it, and public confidence in the movement was impaired. His own account, read at source, contains the words "I have fallen," an acceptance of blame, and a submission to his church's judgment. It is worth reading rather than inferring from its chapter headings, which suggest a man rebutting a charge rather than owning a relapse. He had relapsed once before, twenty-nine months earlier, when his influence weight was small, and that episode was handled internally and quickly. Same man, same illness, same candor: what differed was the weight.

7.3 Maxwell, and a priority problem stated plainly

Milton Maxwell's 1950 comparison of the two fellowships reaches, without any formal apparatus, a substantial part of this paper's conclusion. His final section lists AA's advantages as exclusively alcoholic membership, singleness of purpose, a definite program of recovery, anonymity, and what he calls hazard-avoiding traditions; he writes that a comparison with the Washingtonian experience underscores *the sheer survival value* of the principle of anonymity; and he reaches that conclusion by way of Gough's relapse and what it cost a movement whose credibility sat in named men. He also identifies the tradition of keeping authority in principles rather than in offices and personalities, and connects it to rotating leadership.

What this paper adds to Maxwell is the theorem, and nothing else. He had the observation, the mechanism, and the case. He had no formal condition to which the observation could be referred, and therefore no way to say why anonymity should have survival value rather than merely that it did. Whether the vanishing-influence condition is what Maxwell was pointing at is a separate question, and this paper does not settle it.

7.4 What the case cannot do

It is one case, selected because it is the obvious comparison, and the direction of selection is unfavorable: the Washingtonians are famous among people interested in AA precisely because the contrast is instructive. Nothing here establishes that the missing enabling provisions caused the decline; the movement was also absorbed by a temperance politics it had defined itself against, and the manual's anti-politics article had nothing to bite on once the movement's identity was itself a political position. How widely the Utica model constitution was actually adopted is not recorded in the manual and we have found no source that settles it. The historical material is offered as an existence proof that the distinction between the two tiers of rule is visible in a real code, not as evidence about the consequences of omitting one tier.

8. Limitations

8.1 The largest one, stated first

The mapping in Table 1 is an interpretation of the Traditions' wording, arrived at by the author, and no computation in this paper touches it. Everything the paper claims about AA specifically, as opposed to about stochastic matrices, passes through it. A reader who holds that Tradition 2 concerns humility rather than weighting, or that anonymity is chiefly protective of individuals, can accept every number here and reject the paper's thesis. This is not a caveat on a result; it is the status of the result.

What would settle it is elicitation: give the Traditions' published short text to readers who do not know the hypothesis, ask them to say what each rule constrains, and measure agreement with the mapping. That has not been done. A single reader who knows the rooms did read a

draft and identified a substantive error, the membership-versus-attendance conflation of §6.1, and no other intervention in the project changed as much for as few words. The implication for the parts nobody has checked is uncomfortable and is the reason this limitation is listed first.

8.2 The remaining limitations

1. **No parameter is estimated.** Section 3's computations demonstrate a theorem on constructed matrices; Section 6's are simulations from 118 assumed values, none fitted, because the longitudinal data such a model would need has never been collected. The paper's empirical content is its predictions, not its numbers. A model of this kind can show that a set of ideas is consistent and that a mechanism is available. It cannot show that anything is true.
2. **The S and G matrices are hand-written judgments**, and all of Section 4 plus the resource structure of Section 6 inherits from them. Their tolerance of ± 30 per cent perturbation is necessary and not sufficient support, and the structural randomization they fail (§4.5) is the design that speaks to the question. The correct remedy is a second governance matrix elicited independently from another reader, scored by Cohen's kappa against the first. A blank elicitation form exists; no second reader has completed it. Until one does, Section 4 rests on one person's judgment.
3. **Member heterogeneity was tuned to produce a desired behavior.** Its standard deviation was selected in an attempt to convert individual cliffs into a graded group response. After mean-centring and re-estimating the state-dependent environment, the corrected audit finds typical-member bistability in only 1.75 per cent of 400 endpoint environments. The intended mechanism is therefore not established by the released baseline, and Section 6's quantities inherit from the authored spread.
4. **Specification search.** Several architectural choices (the retention channel for Tradition 3, core-based carrying capacity, the placement of the capacity gate) were refined against simulation behavior as well as substantive reasoning. This risks tailoring an architecture to expected behavior and no assurance can be given that it has not happened. The four structural variants of §6.6 are a partial and inadequate answer.
5. **Thirty-five parameters cannot affect a fully adherent group at all**, because the governance matrix cancels exactly at full adherence. Any sensitivity result quoted over the full 118 must be read with that in mind, and a design that perturbs those 35 and reports no effect has found nothing.
6. **Conditional figures read as unconditional.** Quality in Table 8 is computed among members of surviving groups. Read without the surviving fraction beside it, "quality is maintained" would describe a population that has partly ceased to exist. Table 9 exists to make the difference visible.
7. **DeGroot averaging is a strong simplification of a group conscience.** Real members argue, defer selectively, abstain, update out of order, and sometimes harden rather than converge. Golub and Jackson's result is about naive averaging specifically. Whether real deliberating groups behave like DeGroot updaters has not been tested here or, so far as we have found, anywhere.
8. **Endogeneity of the group input** blocks estimation of Section 5's technology and is unsolved here.
9. **The sign of ρ has never been measured.** The contribution is to make the question answerable, not to answer it.
10. **No representation of who attends to whom.** In Section 6's model no member's state enters another member's growth equation except through group aggregates, so there is no sponsorship, no pairing, no clique, and no re-sorting. The mechanism that produced the

Carrell–Sacerdote–West result is absent by construction (§6.7). Building it is a different model, not a different run, and it is the largest single piece of technical work outstanding.

11. **Durations were never calibrated.** The arrival rate and the churn rate are two of the 118 registered numeric values, and the horizon is a separate design choice. Endpoint orderings have outcome-specific sensitivity records, and the 10-, 20-, 30-, 50- and 100-year checks continue to move. The trajectories are shapes inside the model, not forecasts of how long a real group lasts or evidence of steady state.
12. **Combinations were not tested.** Each decline condition switches one thing off at full adherence elsewhere. Real decline is unlikely to be so tidy and there is no reason to expect the costs to add.
13. **A correspondence is not a cause.** That the Traditions implement a known aggregation criterion does not establish that this is why they were written, or why AA has endured.
14. **The decay rate is uncalibrated and consequential.** δ_0 is second only to p in every sensitivity design. The two estimates read for anything comparable measure skills, not practices: Dinerstein, Megalokonomou, and Yannelis (2022) put the depreciation of teaching skill at 4.3 per cent a year for early-career teachers, and Cohen, Johnston, and Lindner (2023) find no measurable decline in cognitive skills over up to three years of unemployment. Read as skill, $\delta_0 = 0.06$ per week is one to two orders of magnitude too fast; read as practice it is untested. A search of the literature on how fast practices lapse brackets it rather than calibrating it. Edgren, Baretta, and Inauen (2025), the first intensive longitudinal study of habit decay in daily life, find self-reported automaticity settling in a median of 9 to 10 days (range 1 to 65) among the 22 per cent of participants whose fitted trajectory crossed the scale midpoint, all of them deliberately degrading an unwanted habit, an order of magnitude faster than δ_0 and a different object: a habit actively removed, not a practice left unrenewed. Kaskutas, Bond, and Avalos (2009) give the other end from inside AA, with four attendance trajectories over seven years in which participation falls away on a scale of years and abstinence does not follow it down, which also cautions against reading the practice variable as a meeting count. The one-at-a-time screen found that lowering δ_0 by 25 per cent or more reverses the ordering of referral loss against pure attraction loss on final membership, and a 400-seed run paired by common random numbers, over 1,560 weeks at $dt = 0.5$ with full adherence otherwise, confirms it and bounds it. At 25 per cent lower, referral loss leaves the larger group, by 14.97 members [12.73, 17.21], but remains worse on endpoint viability, by 0.2450 [0.2028, 0.2872], and on existence, by 0.1600 [0.1240, 0.1960]; at 50 and 75 per cent lower every referral-loss run is endpoint-viable. A second run on the same design, at 10, 15 and 20 per cent lower, locates the membership reversal between 15 and 20 per cent: the ordering holds at 10.92 members [10.18, 11.66] and 6.35 [5.03, 7.68] respectively, and is reversed at -2.79 [-4.77, -0.81], while endpoint viability and existence are reversed in no paired run at any of the three. The ordering is therefore conditional on practice lapsing at roughly the assumed rate, and at slower rates which loss is worse depends on the outcome scored.
15. **Sources not read.** Alexander (1988) on the class and domestic dimensions of the Washingtonian movement; Blumberg (1980, 1991) on its political entanglement; White's *Slaying the Dragon*. Maxwell (1950) was read in full, but the copy available to us is a retyped web reproduction with visible transcription errors, not a scan of the journal, so every Maxwell citation here has been checked against a transcription rather than against the journal. *Twelve Steps and Twelve Traditions* (1953) was read in full on 10 August 2026. Of the rest of AA's own literature, *Alcoholics Anonymous Comes of Age*, *Pass It On* and all but one 1946 article of the *Grapevine* have not been obtained.

9. Falsifiable Predictions

Where the analysis reported above bears on a prediction, or narrows what it should say, this is marked.

1. **Decision quality improves with group size under flat influence and plateaus under concentrated influence.** Testable with forecast or vignette tasks administered to group consciences of varying structure. Unchanged, and untested.
2. **Rotation breadth, not rotation per se, predicts group durability.** Testable from service rosters. *Narrowed by the analysis above:* the operational threshold is roughly a quarter of the membership in the rotation pool, computed as twenty-six per cent across sizes from 50 to 800 (§3.5). The effect of falling short is a real but moderate permanent cost, not a near-clique regime, and a test should be powered accordingly.
3. **Groups losing exogenous newcomer inflow decline demographically while established practice among selected viable remnants can remain high.** Unlisted meetings, schedule-disadvantaged meetings, and meetings distant from referral sources provide natural variation. *Narrowed by the corrected trajectories:* viable-room membership also falls sharply, so the interior is not warning-free. The testable divergence is between losses in groups and members and the smaller change in practice among those who remain; it must be measured on all three series.
4. **The two supply channels dissociate**: attraction loss produces smaller remnants at year thirty, referral loss produces delayed dissolution, so proximity to treatment facilities should predict survival through low-attraction periods. *Partly answered, and qualified.* Earlier drafts described this as one of exactly two universal simulation results. The corrected analysis reports strict support, ties, reversals, and unresolved screens separately; it does not promote a finite sensitivity suite into a universal claim. Mortality and membership remain distinct estimands.
5. **Tradition 3 has distinct resource-governance and inverse-practice friction paths.** *Bearing of the analysis above:* at 400 paired seeds, combined loss ends at 6.76 members against 17.80, closes 25.0 per cent of groups, and leaves 54.8 per cent endpoint-viable against 98.5 per cent at baseline. Friction loss alone costs 2.96 members [1.85, 4.08]; governance loss alone costs 6.03 [5.04, 7.01]. The earlier predominantly-size interpretation does not survive the corrected model.
6. **If step order does not predict step completion** ($\rho > 0$ throughout), the chain interpretation of the CES reformulation fails. The proxy-averaging exercise in §5.1 neither identifies latent practice nor establishes a required number of instruments.
7. **the two-tier division should be visible to readers who do not know the hypothesis.** Give the published short text of the Traditions to independent readers and ask which of them supply anything a member uses directly and which only protect other rules. The present split is author-coded, not a tier discovered by the model. If independent readers do not reproduce it, the resource layer is encoding the author's prior rather than the text's content.
8. **the mapping itself is testable by elicitation.** See §8.1. This is the prediction whose failure would cost the paper the most.

10. Conclusion

This paper's durable contribution is a correspondence: three of AA's Twelve Traditions, on a reading of their wording that the paper states plainly and does not verify, jointly implement the vanishing-influence condition under which naive collective deliberation aggregates information, a criterion formalized by Golub and Jackson in 2010 and reached by a fellowship of laypeople,

from eleven years of watching groups fail, in 1946. The correspondence yields two corollaries with immediate empirical content, that rotation must scale with the group and that obstruction is a matter of scaling rather than severity, and it reframes anonymity from an ethic of humility to a structural precondition of trustworthy group decision-making.

Around that center, the paper contributes a method: deriving institutional couplings through an explicit resource layer rather than asserting them, which rejects the natural index-pairing conjecture on all twelve of its instances while recovering, unbidden, the functional division of the Traditions and the primacy of unity. It also contributes an accounting of what that method cannot support. Five of the twelve rejections are structurally forced and invisible to the perturbation design that was originally quoted for them; a threshold test that can see them finds them failing by comparable margins; and the central coupling claim survives structural randomization less often than it fails. Those facts are reported here because a derivation whose robustness is asserted rather than measured is worth less than one whose limits are known.

The membership model's principal prediction, that referral-starved groups decline demographically while the practice of their remaining members stays high, so that the decline shows in the size of the room and not in the quality of those who remain, is specific, mechanistically grounded, distinguishable from rival accounts, and testable with records AA's service structure already keeps. In the corrected five-architecture comparison, referral loss is worse than pure attraction loss on existence, endpoint viability and mean final membership. The expanded parameter screens bound that result over their stated ranges; they are not a proof outside them.

Where this line of work should go next is not further modeling. It is measurement: of the mapping, by elicitation from readers who do not know the hypothesis; of the governance matrix, by a second independent elicitation; of rotation breadth against group longevity; of step ordering against step completion; and of the quiet demographic signature of groups that seekers have stopped finding.

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Status note. An entry explicitly marked read at source, read in full, abstract only, cited at a remove, or not read has that current-project status. For entries without an explicit status, the repository does not document whether the full work was read; they are background citations and do not upgrade a load-bearing claim. The complete current status register is `research/SOURCES.md`. The separately supplied corpus was worked through on 9 August 2026. The two journal articles it could not retrieve, Pagano et al. (2004) and Greenfield and Tonigan (2013), were obtained by hand and read in full on 10 August 2026, and what remains under `research/staged/` is the acquisition report and its metadata, which are provenance rather than evidence. No source document is committed to the repository: each source is published as a citation, a rights position, a provenance URL, a SHA-256 and a vocabulary-only verification index.

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What was not read. Any work testing whether real deliberating groups behave like DeGroot updaters. Any literature on peer-group composition in voluntary mutual-aid settings specifically. Alexander (1988), Blumberg (1980, 1991), and White's *Slaying the Dragon.* Jellinek's per-capita consumption estimates, which are quoted through Maxwell and whose original has not been traced. Any measurement of how fast a practice lapses when nothing renews it: the literature on habit and practice lapse was searched in September 2026 and the two papers that bracket the decay rate were read in full, but neither measures it.

A. The Whole Paper in Plain Language

This appendix says everything the paper says, without the math. It is written for a reader with no background in economics or statistics. Where a natural way of putting something would be wrong, this appendix says so.

A.1 What this paper is about

AA runs on two sets of twelve ideas. The Twelve Steps are for the person. The Twelve Traditions are for the group. This paper turns both into math and checks whether the pieces fit together, and along the way it finds that some things everybody assumes about them are wrong, and some things nobody says about them are true.

A.2 The big finding: why "nobody's in charge" actually works

Think about how an AA group makes a decision. There is no boss. People talk it over until the room agrees. That is called the group conscience.

Now here is a question: when can you trust a decision made that way? In 2010, two economists proved the answer. A group that decides by talking it out can be trusted only if no single person's opinion carries a big fixed chunk of the final answer. Here is why. Everyone's opinion is partly right and partly mistaken. When lots of opinions get blended evenly, the mistakes point in different directions and cancel out, and the more people the better. But if one person's opinion always makes up a third of the result, their mistakes never cancel. Adding more people does not help. The group just gets more and more sure of an answer that is no more likely to be right.

Now look at three of the Traditions. Leaders serve the group; they do not run it. There is no ladder to climb, because service jobs rotate. And everyone is anonymous: no last names, no job titles, no status. Each of those rules does the same thing from a different angle: it stops any one person's voice from getting too heavy. This paper proposes that those mechanisms can reduce stationary influence concentration, which is the condition in the 2010 proof. The wording alone does not establish that they do.

> *The rules written in 1946 are consistent with a condition mathematicians would formalize 64 years later. That is a present-day mapping, not evidence that AA's drafters discovered, intended, or implemented the theorem.*

One honest warning about that paragraph, and it is the most important sentence in this appendix. The proof is real and we have read it. But the step from "Tradition 2 says leaders do not govern" to "therefore nobody's opinion carries extra weight in the math" is our reading of a sentence. Nobody has checked it. None of the computer work in this paper tests it. Somebody could reasonably say those Traditions are about humility, not about arithmetic, and this paper has no answer for them yet. Everything else here is downstream of that one unverified step.

A.3 What it looks like when the rule breaks

Two familiar characters break it. The first is the old-timer whose opinion ends every discussion, not because there is a vote, but because everyone waits to hear what they think. The second is the little circle of long-timers who have already talked it over before the meeting starts.

Here is the alarming part from the math: when this happens, nothing looks wrong. The group still reaches agreement. The meetings feel fine. There is no argument and no drama. The group is simply wrong more often, it is wrong confidently, and getting bigger does not fix it. A broken group conscience feels exactly like a working one from the inside.

There is a third case, worse than both, that we did not expect. If a small group of people not only receives most of the attention but gives all of *their* attention only to each other, then in the long run the rest of the room counts for literally nothing. The group's judgment becomes exactly the judgment of that handful, no matter how many other people are in the chairs.

And it does not take a monster to cause the problem. One person holding five per cent of the room's attention, which is not very much, already nearly doubles how wrong a large group ends up.

A.4 Something the Traditions forgot to mention

The Traditions say to rotate service jobs so nobody becomes a boss. The math agrees, and adds a catch nobody wrote down. Rotating only works if lots of different people take turns. Picture a group of 400 where the same twelve people trade the jobs among themselves forever. Rotation like that leaves a permanent floor under how much attention the rotating dozen hold, and the floor does not go down as the group grows. To really work, roughly a quarter of the group needs to be in the rotation. Any group could check this against its own service list.

One thing not to overstate. It would be easy to say that a group of 400 rotating twelve people behaves almost the same as a group run by a clique. That is too strong. It sits in between, and it is closer to the healthy case than to the clique. The honest description is a real but moderate permanent cost.

And a bigger lesson underneath it. The obvious question to ask about a prominent member is "how prominent?" That is the wrong question. It is "does the prominence grow when the group grows?" A clique that keeps to itself in a room of ten but gets diluted as the room fills is not a problem at all in the long run. A single person who stays equally prominent no matter how big the room gets is a problem forever. Severity is not the test. Scaling is the test.

A.5 The steps don't match the traditions by number

It is tempting to think Step 1 goes with Tradition 1, Step 2 with Tradition 2, and so on down the line. It sounds neat. But it does not hold up, and it was never AA teaching. The Traditions were written about ten years after the Steps, to fix real problems groups were having.

Instead of matching them by number, we asked a different question for each step: what does a person actually need from the group to do this step? Then the matching takes care of itself. Step 5 is telling another human being the worst things you have ever done. What do you need for that? You need to know it will not leave the room. That is anonymity, Tradition 12, not Tradition 5. Step 12 is carrying the message to the alcoholic who still suffers. Tradition 5 says the group's one job is to carry its message to the alcoholic who still suffers. Same sentence, basically. Those two go together.

And when we code this for all twelve steps, two things follow from those authored choices. First, five of the Traditions (autonomy, no endorsements, paying your own way, no hierarchy, no outside opinions) turn out to give members nothing directly. Their whole job is to protect the other seven from being eroded. Those five empty rows were put into the table by the author;

multiplication did not discover them independently. Second, unity has the largest semantic overlap under the same coding. That is a concise description of the table, not a fact about AA groups established by data.

Now the part that argues against the finding. We tested how much of this survives if you disagree with the numbers we made up.

Unity's importance holds up well. Even if you throw away every number we chose and keep only the pattern of which rule touches which need, unity still comes out on top about three times in four.

The rest does not hold up nearly as well. Do the same thing to the "steps-don't-match-traditions" finding and it only survives about four times in ten, which is less often than it fails. Worse, five of our twelve pieces of evidence for it were not really evidence at all: those five Traditions had zeros in our table by construction, so of course they never matched. We built a different test that could actually see those five, and they still fail, by about the same margins as the other seven. So we think the finding is right. But it rests on our judgment about the numbers, not on a proof, and anyone who thinks we chose the numbers badly is entitled to reject it. That is worth saying plainly rather than dressing it up.

A.6 Can you skip a step? Nobody has ever checked

Everyone in AA says you cannot skip a step, that you cannot make honest amends (Step 9) for harms you never wrote down (Step 4). It sounds obviously true. But as far as we could find, no researcher has ever actually tested whether people work the steps in order, or whether doing an early step really is what makes a later one possible.

This paper turns that belief into one measurable number. If the number comes out one way, the steps are truly a chain: skip a link and everything after it fails. If it comes out the other way, they are more like a menu, and a strong group can carry someone past a step they missed. The number might be measured with a purpose-built longitudinal design. Our limited exercise adds noise only to an output while supplying the inputs and loadings without error; it shows that averaging more output proxies can improve resolution. It does not establish how many instruments a valid latent-variable design needs.

A.7 What can actually go wrong for a group

Start with a fact about AA that shapes everything here: the only requirement for *membership* is a desire to stop drinking, and no group can take that away from anybody.

One thing that is easy to get wrong here. It is tempting to put this as "a group cannot close its doors," and that is not right. Closed meetings are ordinary and long-standing, and a closed meeting is about who is in the room, not about who is a member. Nobody can be thrown out of AA. Somebody can absolutely be in a room where they are not welcome. In the executable model Tradition 3 affects both resource governance and inverse-practice-weighted dropout friction; it does not record tenure and therefore cannot identify literal newcomer retention. The two paths are now tested separately.

So if a group cannot revoke anyone's membership, what can actually go wrong? Three different things, and they leave three different marks:

1. **The group becomes invisible.** Nobody finds it anymore: it is not listed, it meets at a bad time, word of mouth dried up. The model says this group is much smaller at year thirty, kept alive by court cards and treatment-center referrals: about seven tenths of a healthy group's size (12.4 members against 17.8). It does not die, and the practice of the people in it is at most slightly lower than in a healthy group, by 0.0067 at year ten, smaller than the half-widths of the two estimates.

2. **The referrals dry up.** The treatment center closed; the court program ended. The group coasts on word of mouth and shrinks. At year ten about one group in three has already dropped to five members or fewer, and by year thirty nine in ten have closed altogether (89.5 per cent), leaving a handful of survivors.
3. **The group gets unwelcoming.** Nobody loses their membership, but there is an in-crowd, and people who come to it can feel it, and those with little practice drop out sooner. Averaged over every run, this group is under two fifths the size of a healthy one at year thirty (6.8 members against 17.8). One group in four closes outright and a little over half are still viable at the end.

How much you can trust any of these numbers depends on how many times the simulation was run. Every figure here comes from four hundred runs paired by common random numbers, and each carries an interval. The project's own history is the warning. An earlier version of the model put a healthy group at about 42 members; after a correction to how member capability is drawn, and with four hundred runs, it is 17.8. A number produced by a random simulation and quoted to three digits from a handful of runs is not a result at all, and it is very easy to print one without noticing.

And here is the strangest thing the model says. In the case that actually kills groups, the people still in the room are doing fine. Their practice holds up right to the end, and if anything looks slightly *better* than average, because the ones who were struggling have already gone. Judged by how the remaining members are doing, that dying group looks like a healthy one. If that is true in the real world, and it is checkable, it means the quality of the remaining members is a poor guide to whether the group is in trouble. The size of the room is a better one: how many newcomers came this year, and how many came back.

One qualification, because the loose version of this is wrong. This is selection, not health, and the room does give a warning. Among the groups still viable, membership falls from about 22 at year five to about 11 at year twenty. The claim that nothing inside a dying group warns of its decline is not supported, and it is withdrawn: what stays high is the practice of the people who remain, not the size of the group they remain in.

A.8 Does it matter who starts a group?

We tested three founding-state distributions with the same mean initial practice. Their paired final-membership differences are imprecise and no equivalence margin was specified. The correct result is unresolved, not that all three are the same.

You should not believe that result very much, and the reason is a famous experiment. Researchers at the Air Force Academy measured how much cadets help each other study, then used those measurements to build squadrons designed to help the weakest students. Their own model predicted a gain. What actually happened was a loss of about the same size, on exactly the students they had tried to help. The reason was that once you put fifteen strong students and fifteen weak ones in a room with nobody in between, the weak ones stopped mixing with the strong ones and found each other instead. The measurement was real; building a room out of it destroyed the thing that had been measured.

Our model has no way for people to choose who they spend time with, so it could not reproduce the Air Force mechanism. That study is a warning about transport and endogenous mixing, not a measurement of the upside or downside of arranging an AA group. No causal claim about AA composition follows from this comparison.

A.9 The movement that came before, and what it actually did

AA was not the first fellowship of drunks helping each other stay sober. In 1840, six men in a Baltimore tavern founded the Washingtonians. It spread across the country in four years and

was essentially finished in ten.

The story usually told, especially in AA circles, is that they died of having no rules. **We checked, and that story is false.** Their own handbook from 1842 turned up, and it contains written rules: each society independent and answering to nobody above it, funds controlled by its own members, and nothing political or religious allowed into the meetings. They passed a model constitution at a convention in February 1842, printed it in a newspaper, reprinted it because of demand, and told every new society to get a copy.

What they did not have was any of the rules on the other side of the ledger, the ones that give a member something. And on anonymity they took the opposite view on purpose, with a reason: their handbook tells new joiners to stand up and call out their names, because publicity is better than "whisperings and secresy."

So the real comparison is not rules against no rules. It is one written code against another, and the ones missing from the first are exactly the ones the math points at. That is a better comparison than the one we started with, and it is better because it is true.

Their most famous speaker relapsed in public in 1845, opponents made the most of it, and the movement's credibility suffered. He had relapsed once before, quietly, two and a half years earlier, and the institutional response was much smaller. The same-person contrast is suggestive, but it does not identify influence weight as the sole cause and the first episode was not literally costless.

A.10 How much of this should you believe?

Here is the honest answer, and it is less flattering than the summary above may have sounded.

The mathematical theorem in A.2 is real and settled. The claim that AA's three rules are the same thing as the theorem's condition is *our reading of three sentences*, and nobody has checked it. That is the whole hinge of the paper and it is the part with the least support.

The finding about rotation follows from the theorem and is solid math.

The step-and-tradition results come from two tables of numbers we made up, and they do not survive if you throw those numbers away. We say so in the paper rather than quoting a robustness figure that belongs to a different claim.

Everything from the computer model has never been checked against real people or real groups. Every number in it was chosen by us, not measured. The sensitivity exercises are screens over specified ranges, not proofs that claims rest on no number. Their strict, tied, and reversed outcomes must be reported separately, and the earlier referral comparison used a mixed Tradition 11 intervention. The corrected headline compares referral loss with the pure attraction path, while structural size and endpoint-viability orderings are reported separately.

A model like that can show that ideas fit together. It cannot show they are true.

Treat the group-decision math as settled mathematics, treat its application to AA as an unverified reading, and treat everything else as sharp questions waiting for someone to check them. Be suspicious of any idea, ours included, that feels certain before it has been tested. The claims here that felt most obvious at the outset are, without exception, the ones that needed the most narrowing.

> One thing this paper must never be used for: judging any individual person's recovery. It has nothing to say about whether you, or anyone, is "doing it right." It has never been tested on a single human being. For anything personal, talk to a sponsor, a doctor, or a counselor.

A.11 The one-sentence version

> *The rule that nobody in AA is in charge and nobody uses their last name may not be just humility: read a certain way, it is the exact mathematical condition that makes a group's decisions trustworthy, written down by people who learned it from failure*

sixty-four years before anyone proved it. Whether that reading is right is the thing still to be checked.

B. Reproducibility Note

Every quantitative claim in this paper is reproducible from the companion repository without reference to any external dataset, because there is no external dataset.

Deterministic results. Section 3 and the coupling derivation of Section 4 are algebra on constructed matrices. Influence vectors are normalized left dominant eigenvectors; consensus errors come from the closed form in equation (1); the coupling is a matrix product; the threshold test is one division per Step. These carry no sampling error and are exact to the digits printed. The magnitude-randomization and sparsity designs of Section 4 are not of this kind: they are Monte Carlo over 2,000 draws each and carry Wilson intervals. The 1,000-draw executable-model screen in the same section is a three-seed screen, not a confirmatory contrast.

Stochastic results. Section 6 reports Monte Carlo output. The design is: Euler integration at $dt = 0.5$ weeks over 1,560 weeks, except that the paired Tradition comparisons run 1,040 weeks; **400 seeds for every confirmatory contrast**, with the 200-seed numerical-resolution and horizon checks and the three- or five-seed parameter screens labelled as what they are; survival reported with Wilson intervals; membership and quality with 95 per cent half-widths from the cross-run standard error; Tradition comparisons under common random numbers at 400 paired replications. **Common random numbers reduce variance and do not reduce the replication count required**: the comparison in Table 10 was originally run at 30 paired replications, produced a different ranking, and reversed when rerun at 400. Numerical checks use 200 paired seeds at $dt = 1.0$, 0.5, 0.25 and 0.125 weeks and 200 seeds at 10, 20, 50 and 100 years. These support the finite-horizon numerical resolution and show continued long-run movement; they do not prove a steady state.

Perturbation designs. Multiplicative jitter at $L \in \{0.15, 0.30, 0.50, 0.75\}$, 2,000 draws per level, seed 3. Structural randomization replacing every non-zero entry with a uniform draw on $[0.05, 1.00]$, 2,000 draws, seed 23. Wilson intervals throughout on $n = 2{,}000$. A multiplicative design cannot move a structural zero and is not evidence about sparsity; the structural design is not evidence about where the zeros belong.

Named threats. Selection effects in the historical case (§7.4); conditioning on survival, addressed by printing the surviving fraction beside every conditional quality figure; specification search on architecture (§8.2); the exact cancellation of 35 governance parameters at full adherence; and, largest of all, the unverified mapping of §3.2.

Verification. All figures are asserted against recomputed values in a notebook executed end to end in a single process, which exits non-zero on any exception, any failed assertion, or any cell producing no output. A separate checker verifies consistency *between* documents: that every decimal is reachable from the notebook, that simulation output states its seeds and an interval, and that no citation attributes to a source something that source does not contain. Each of those checks exists because an error of exactly that kind got as far as a written page before it was caught.